TabuLLM: AI Benchmark Report

1. Executive Summary: Consolidated Audit Results

This report consolidates the results of the TabuLLM data augmentation pipeline. The objective is to audit the fidelity, diversity, and privacy of the synthetic data generated via LLM inference. Below are the averaged performance indicators across all processed datasets in this session.

Audit Indicator	Global Session Mean (± Std)
Synthetic Fidelity (JSD)	76.8 ± 1.40
Correlation Stability	84.7 ± 2.00
Distribution Consistency	84.5 ± 2.25
Entropy Preservation	87.3 ± 1.49
Mean Z Score Drift	0.2 ± 0.03
Categorical Fidelity	83.9 ± 0.54
Privacy Risk	49.6 ± 1.11
Data Density	100.0 ± 0.01
Coverage Score	15.1 ± 5.15
Feature Accuracy	77.8 ± 2.40
Novelty Score	0.4 ± 0.01

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Dataset Audit: indian_liver_patient_dataset.csv

1. Dataset Portrait

Attribute	Value
Rows	583
Columns	11
Numeric features	10
Categorical features	1
Missing cells (%)	0.1%

2. Generation Strategy & LLM Config [1]

Parameter	Configuration
Algorithm (*12)	gpt-4o
Folds (*13)	10
Context % (*14)	20%
Generation % (*15)	100%
Join context (*16)	True

3. Statistical Significance & Data Drift

Feature	Orig Mean	Aug Mean	Shift	P-Value	Status
Age	44.746	44.006	-0.740	1.7%	OK
TB	3.299	1.257	-2.042	61.9%	Drift
DB	1.486	0.477	-1.009	67.9%	Drift
Alkphos	290.576	214.824	-75.752	26.1%	Drift
Sgpt	80.714	38.695	-42.018	52.1%	Drift
Sgot	109.911	44.921	-64.990	59.1%	Drift
TP	6.483	6.729	+0.246	3.8%	OK
ALB	3.142	3.358	+0.216	6.9%	OK
A/G Ratio	0.947	1.005	+0.057	6.1%	OK
Selector	1.286	1.410	+0.123	9.6%	OK

4. Quality & Fidelity Metrics (Mean ± Std)

Metric Category	Aggregated Score
Synthetic Fidelity (JSD) (*1)	76.8 ± 1.40
Correlation Stability (*2)	84.7 ± 2.00
Distribution Consistency (*3)	84.5 ± 2.25
Entropy Preservation (*4)	87.3 ± 1.49
Mean Z Score Drift (*5)	0.2 ± 0.03
Categorical Fidelity (*6)	83.9 ± 0.54
Privacy Risk (*7)	49.6 ± 1.11
Data Density (*8)	100.0 ± 0.01
Coverage Score (*9)	15.1 ± 5.15

Feature Accuracy (*10)	77.8 ± 2.40
Novelty Score (*11)	0.4 ± 0.01

*[1] (*12) Algorithm: Contextual constraint or model parameter.*

*[2] (*13) Folds: Contextual constraint or model parameter.*

*[3] (*14) Context %: Contextual constraint or model parameter.*

*[4] (*15) Generation %: Contextual constraint or model parameter.*

*[5] (*16) Join context: Contextual constraint or model parameter.*

[6] Synthetic Fidelity (JSD): Overall distribution similarity.
[7] Correlation Stability: Multivariate relationships preserved.
[8] Distribution Consistency: Median feature similarity.
[9] Entropy Preservation: Information diversity maintained.
[10] Mean Z Score Drift: Standardized distribution shift.
[11] Categorical Fidelity: Category distribution similarity.
[12] Privacy Risk: Synthetic rows too close to originals.
[13] Data Density: Non-missing cells in augmented data.
[14] Coverage Score: Synthetic coverage of original space.
[15] Feature Accuracy: Mean feature similarity.
[16] Novelty Score: Distance ratio to nearest neighbors.

6. Downstream ML Utility: Comparative Analysis

Model: KNN

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.6863247863247863	0.6871794871794872	0.7017094017094017	0.015384615384615396	
F1	0.7971299319768493	0.7672777765603028	0.7933135982163433	-0.003816333760506163	

Model: DT

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.6606837606837608	0.5760683760683765	0.6794871794871795	0.01880341880341892	
F1	0.7636733306573624	0.6739082378054387	0.7790696048385286	0.015396274181166014	

Model: GB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.6820512820512821	0.6170940170940171	0.7	0.01794871794871794	
F1	0.7889575145067234	0.7106314854125871	0.7942929750466778	0.005335460539954495	

Model: SVM

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.7068376068376068	0.6683760683760684	0.6965811965811965	-0.010256410256410243	
F1	0.8279091624529537	0.7421746178621875	0.7895093165698508	-0.038399845883102836	*

Model: MLP

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.6854700854700855	0.6504273504273504	0.7145299145299145	0.029059829059829057	
F1	0.7981802099960874	0.7329488499448276	0.8127039230836512	0.014523713087563728	

Model: LR

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.7034188034188034	0.6521367521367522	0.6940170940170949	-0.009401709401709379	
F1	0.8108734553482932	0.7368286940163772	0.7922148624759563	-0.01865859287233692	*

Model: XGB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.7025641025641025	0.6435897435897436	0.7145299145299144	0.011965811965811968	
F1	0.7997903571549948	0.7386133684693829	0.8038339051311929	0.004043547976198348	

Model: RF

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.7102564102564103	0.6606837606837608	0.7230769230769231	0.01282051282051282	
F1	0.8128968020770919	0.7545694583658294	0.8173861101140382	0.004489308036946349	

Model: AB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.6743589743589744	0.6777777777777778	0.6974358974358974	0.023076923076923085	
F1	0.7821343746683586	0.7701648366211551	0.7890353846678431	0.006901009999484298	

Model: NB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.5589743589743599	0.7059829059829059	0.6076923076923076	0.04871794871794871	*
F1	0.5587432119380917	0.7870324850778888	0.6378117622287750	0.07906855029068333	*

Appendix: Benchmark Data Trace

Legend: [Blue] = Real Context | [Green] = Generated Synthetic | [Yellow] = Real Training (Remaining) | [Grey] = Real Test Baseline

A. Baseline: Original Dataset (Full Record Sample)

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	65	Female	0.7	0.1	187	16	18	6.8	3.3	0.9
2	62	Male	10.9	5.5	699	64	100	7.5	3.2	0.74
3	62	Male	7.3	4.1	490	60	68	7.0	3.3	0.89
4	58	Male	1.0	0.4	182	14	20	6.8	3.4	1.0
5	72	Male	3.9	2.0	195	27	59	7.3	2.4	0.4
6	46	Male	1.8	0.7	208	19	14	7.6	4.4	1.3
7	26	Female	0.9	0.2	154	16	12	7.0	3.5	1.0
8	29	Female	0.9	0.3	202	14	11	6.7	3.6	1.1
9	17	Male	0.9	0.3	202	22	19	7.4	4.1	1.2
10	55	Male	0.7	0.2	290	53	58	6.8	3.4	1.0
11	57	Male	0.6	0.1	210	51	59	5.9	2.7	0.8
12	72	Male	2.7	1.3	260	31	56	7.4	3.0	0.6
13	64	Male	0.9	0.3	310	61	58	7.0	3.4	0.9
14	74	Female	1.1	0.4	214	22	30	8.1	4.1	1.0
15	61	Male	0.7	0.2	145	53	41	5.8	2.7	0.87
16	25	Male	0.6	0.1	183	91	53	5.5	2.3	0.7
17	38	Male	1.8	0.8	342	168	441	7.6	4.4	1.3
18	33	Male	1.6	0.5	165	15	23	7.3	3.5	0.92
19	40	Female	0.9	0.3	293	232	245	6.8	3.1	0.8
20	40	Female	0.9	0.3	293	232	245	6.8	3.1	0.8

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B.1 Audit Trace: Fold 1 | Ctx: 20% | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	29	Male	1.0	0.3	75	25	26	5.1	2.9	1.3
2	35	Female	1.0	0.3	805	133	103	7.9	3.3	0.7
3	42	Female	7.4	3.6	298	52	102	4.6	1.9	0.7
4	4	Male	0.8	0.2	460	152	231	6.5	3.2	0.9
5	21	Male	18.5	9.5	380	390	500	8.2	4.1	1.0
6	46	Male	0.7	0.2	224	40	23	7.1	3.0	0.7
7	34	Male	1.1	0.4	210	45	50	6.5	3.2	1.0
8	28	Female	0.7	0.2	150	20	18	7.0	3.8	1.2
9	45	Male	2.2	1.1	320	60	70	5.8	2.5	0.8
10	52	Female	0.9	0.3	180	25	30	6.9	3.4	1.1
11	39	Male	1.5	0.6	250	55	65	5.5	2.7	0.9
12	31	Female	0.8	0.2	160	22	20	7.2	3.6	1.0
13	33	Male	7.1	3.7	196	622	497	6.9	3.6	1.09
14	38	Male	2.2	1.0	310	119	42	7.9	4.1	1.0
15	64	Male	3.0	1.4	248	46	40	6.5	3.2	0.9
16	40	Female	0.9	0.2	285	32	27	7.7	3.5	0.8

B.2 Audit Trace: Fold 2 | Ctx: 20% | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	34	Female	0.6	0.1	161	15	19	6.6	3.4	1.0
2	90	Male	1.1	0.3	215	46	134	6.9	3.0	0.7
3	60	Male	5.2	2.4	168	126	202	6.8	2.9	0.7
4	26	Female	0.9	0.2	154	16	12	7.0	3.5	1.0
5	45	Male	2.3	1.3	282	132	368	7.3	4.0	1.2
6	36	Male	1.7	0.5	205	36	34	7.1	3.9	1.2
7	29	Female	0.7	0.2	180	25	30	6.8	3.5	1.1
8	52	Male	1.2	0.4	210	40	50	7.0	3.2	0.9
9	41	Female	0.9	0.3	190	30	35	6.5	3.0	1.0
10	33	Male	1.5	0.5	220	50	60	7.2	3.4	1.2
11	47	Female	0.8	0.2	170	20	25	6.9	3.6	1.3
12	55	Male	2.0	0.8	240	60	70	6.7	3.1	0.8
13	50	Female	1.0	0.3	191	22	31	7.8	4.0	1.0
14	52	Female	0.6	0.1	194	10	12	6.9	3.3	0.9
15	30	Male	0.8	0.2	182	46	57	7.8	4.3	1.2
16	58	Male	0.8	0.2	298	33	59	6.2	3.1	1.0

B.3 Audit Trace: Fold 3 | Ctx: 20% | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	58	Male	1.0	0.4	182	14	20	6.8	3.4	1.0
2	42	Male	6.8	3.2	630	25	47	6.1	2.3	0.6
3	72	Male	1.7	0.8	200	28	37	6.2	3.0	0.93
4	38	Female	2.6	1.2	410	59	57	5.6	3.0	0.8
5	25	Female	0.7	0.1	140	32	25	7.6	4.3	1.3
6	62	Male	0.6	0.1	160	42	110	4.9	2.6	1.1
7	45	Female	1.2	0.5	210	30	35	6.5	3.2	1.0
8	60	Male	2.5	1.1	250	40	50	6.8	3.0	0.9
9	34	Male	0.8	0.3	180	25	20	7.1	4.0	1.2
10	52	Female	1.5	0.7	220	35	45	6.0	2.8	0.8
11	29	Male	0.7	0.2	160	18	22	7.3	3.9	1.3
12	67	Female	2.0	0.9	230	42	60	5.9	2.6	0.7
13	44	Female	1.9	0.6	298	378	602	6.6	3.3	1.0
14	30	Male	0.8	0.2	182	46	57	7.8	4.3	1.2
15	53	Male	0.9	0.4	238	17	14	6.6	2.9	0.8
16	28	Male	0.8	0.3	190	20	14	4.1	2.4	1.4

B.4 Audit Trace: Fold 4 | Ctx: 20% | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	42	Male	6.8	3.2	630	25	47	6.1	2.3	0.6
2	60	Male	2.0	0.8	190	45	40	6.0	2.8	0.8
3	49	Female	0.6	0.1	185	17	26	6.6	2.9	0.7
4	37	Male	1.8	0.8	215	53	58	6.4	3.8	1.4
5	38	Male	1.0	0.3	216	21	24	7.3	4.4	1.5
6	48	Male	0.7	0.1	1630	74	149	5.3	2.0	0.6
7	35	Female	1.2	0.5	210	30	40	6.8	3.5	1.1
8	42	Male	0.9	0.2	180	25	30	7.1	3.8	1.2
9	29	Female	0.7	0.1	150	20	25	6.5	3.2	1.0
10	50	Male	2.5	1.0	300	45	60	7.0	3.0	0.9
11	38	Female	1.0	0.4	190	22	28	6.9	3.4	1.1
12	45	Male	3.0	1.5	250	50	70	6.7	3.1	0.8
13	42	Male	0.7	0.2	197	64	33	5.8	2.4	0.7
14	69	Male	0.9	0.2	215	32	24	6.9	3.0	0.7
15	53	Male	1.6	0.9	178	44	59	6.5	3.9	1.5
16	49	Male	1.3	0.4	206	30	25	6.0	3.1	1.06

B.5 Audit Trace: Fold 5 | Ctx: 20% | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	32	Male	15.0	8.2	289	58	80	5.3	2.2	0.7
2	72	Male	0.7	0.2	185	16	22	7.3	3.7	1.0
3	50	Male	0.7	0.2	192	18	15	7.4	4.2	1.3
4	72	Male	0.7	0.1	196	20	35	5.8	2.0	0.5
5	20	Male	1.1	0.5	128	20	30	3.9	1.9	0.95
6	65	Female	0.7	0.2	406	24	45	7.2	3.5	0.9
7	45	Male	1.2	0.4	210	45	50	6.8	3.2	1.0
8	34	Female	0.9	0.3	190	30	40	7.1	3.5	1.1
9	60	Male	2.5	1.0	300	60	70	5.9	2.8	0.9
10	29	Female	0.8	0.2	180	25	30	7.5	3.9	1.2
11	55	Male	3.0	1.5	400	80	90	6.0	2.5	0.8
12	42	Female	1.0	0.3	220	35	45	7.2	3.6	1.0
13	33	Male	2.1	0.7	205	50	38	6.8	3.0	0.7
14	22	Female	6.7	3.2	850	154	248	6.2	2.8	0.8
15	48	Male	5.8	2.5	802	133	88	6.0	2.8	0.8
16	84	Female	0.7	0.2	188	13	21	6.0	3.2	1.1

B.6 Audit Trace: Fold 6 | Ctx: 20% | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	18	Male	1.3	0.7	316	10	21	6.0	2.1	0.5
2	26	Female	0.6	0.2	142	12	32	5.7	2.4	0.75
3	18	Male	0.9	0.3	300	30	48	8.0	4.0	1.0
4	40	Female	0.9	0.2	285	32	27	7.7	3.5	0.8
5	27	Male	0.7	0.2	243	21	23	5.3	2.3	0.7
6	43	Male	1.3	0.6	155	15	20	8.0	4.0	1.0
7	34	Male	0.9	0.3	210	45	50	7.2	3.5	1.0
8	29	Female	1.1	0.4	180	30	40	6.8	3.2	0.9
9	42	Male	0.8	0.2	190	25	35	7.0	3.8	1.2
10	37	Female	1.3	0.6	220	50	60	6.5	3.0	0.8
11	50	Male	2.0	1.0	250	60	70	7.5	3.9	1.1
12	28	Female	0.7	0.2	160	20	25	6.9	3.4	1.0
13	50	Male	0.9	0.3	194	190	73	7.5	3.9	1.0
14	49	Male	0.6	0.1	218	50	53	5.0	2.4	0.9
15	18	Male	0.8	0.2	228	55	54	6.9	4.0	1.3
16	30	Female	0.7	0.2	194	32	36	7.5	3.6	0.92

B.7 Audit Trace: Fold 7 | Ctx: 20% | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	18	Male	0.6	0.2	538	33	34	7.5	3.2	0.7
2	66	Female	2.9	1.3	168	21	38	5.5	1.8	0.4
3	73	Male	1.9	0.7	1750	102	141	5.5	2.0	0.5
4	46	Male	18.4	8.5	450	119	230	7.5	3.3	0.7
5	26	Male	6.8	3.2	140	37	19	3.6	0.9	0.3
6	42	Male	0.8	0.2	114	21	23	7.0	3.0	0.7
7	34	Male	1.2	0.5	210	45	60	6.8	3.2	1.0
8	29	Female	0.8	0.2	150	20	25	7.0	3.5	1.2
9	45	Male	3.5	1.8	320	70	90	5.5	2.5	0.8
10	52	Female	0.9	0.3	180	30	40	6.5	3.0	1.1
11	61	Male	2.1	1.0	250	50	80	6.0	2.8	0.9
12	37	Female	0.7	0.2	160	22	30	7.2	3.6	1.3
13	13	Female	0.7	0.2	350	17	24	7.4	4.0	1.1
14	54	Female	23.2	12.6	574	43	47	7.2	3.5	0.9
15	63	Male	0.9	0.2	194	52	45	6.0	3.9	1.85
16	46	Male	0.6	0.2	290	26	21	6.0	3.0	1.0

B.8 Audit Trace: Fold 8 | Ctx: 20% | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	45	Female	0.8	0.2	165	22	18	8.2	4.1	1.0
2	32	Male	22.7	10.2	290	322	113	6.6	2.8	0.7
3	66	Female	4.2	2.1	159	15	30	7.1	2.2	0.4
4	43	Female	0.9	0.3	140	12	29	7.4	3.5	1.8
5	60	Male	2.1	1.0	191	114	247	4.0	1.6	0.6
6	39	Male	1.6	0.8	230	88	74	8.0	4.0	1.0
7	52	Female	1.2	0.4	180	30	40	7.2	3.8	1.1
8	29	Male	2.3	1.0	210	45	65	6.5	3.0	0.9
9	61	Male	0.9	0.2	160	20	25	6.8	3.5	1.2
10	47	Female	1.5	0.6	220	50	70	7.0	3.6	1.0
11	33	Male	3.0	1.4	250	60	80	5.9	2.9	0.8
12	55	Female	0.8	0.2	140	18	22	7.5	4.0	1.3
13	48	Male	0.7	0.2	208	15	30	4.6	2.1	0.8
14	60	Male	22.8	12.6	962	53	41	6.9	3.3	0.9
15	27	Male	1.0	0.3	180	56	111	6.8	3.9	1.85
16	28	Female	0.8	0.2	309	55	23	6.8	4.1	1.51

B.9 Audit Trace: Fold 9 | Ctx: 20% | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	48	Male	0.7	0.1	1630	74	149	5.3	2.0	0.6
2	60	Male	5.7	2.8	214	412	850	7.3	3.2	0.78
3	33	Male	0.8	0.2	135	30	29	7.2	4.4	1.5
4	40	Male	30.8	18.3	285	110	186	7.9	2.7	0.5
5	26	Male	1.0	0.3	163	48	71	7.1	3.7	1.0
6	27	Male	0.7	0.2	243	21	23	5.3	2.3	0.7
7	45	Male	1.2	0.4	210	55	60	6.8	3.2	1.0
8	30	Female	0.7	0.2	145	20	25	7.0	4.0	1.2
9	55	Male	3.5	1.5	250	70	120	6.5	2.5	0.8
10	40	Female	1.0	0.3	180	30	40	7.2	3.8	1.1
11	60	Male	2.0	0.8	220	45	50	6.0	2.9	0.9
12	35	Female	0.8	0.2	160	25	30	7.5	4.2	1.3
13	26	Male	2.0	0.9	195	24	65	7.8	4.3	1.2
14	47	Male	2.7	1.3	275	123	73	6.2	3.3	1.1
15	74	Male	1.0	0.3	175	30	32	6.4	3.4	1.1
16	48	Male	3.2	1.6	257	33	116	5.7	2.2	0.62

B.10 Audit Trace: Fold 10 | Ctx: 20% | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	45	Male	2.3	1.3	282	132	368	7.3	4.0	1.2
2	21	Male	0.8	0.2	183	33	57	6.8	3.5	1.0
3	31	Female	0.8	0.2	215	15	21	7.6	4.0	1.1
4	51	Male	2.9	1.3	482	22	34	7.0	2.4	0.5
5	34	Male	8.7	4.0	298	58	138	5.8	2.4	0.7
6	39	Male	0.6	0.2	188	28	43	8.1	3.3	0.6
7	47	Male	1.2	0.4	210	45	60	6.8	3.2	0.9
8	30	Female	0.7	0.2	180	20	25	7.0	3.8	1.2
9	55	Male	2.5	1.1	320	50	75	5.9	2.6	0.8
10	40	Female	0.9	0.3	190	30	40	7.5	4.1	1.1
11	62	Male	1.8	0.7	250	60	90	6.0	2.9	0.7
12	29	Female	0.6	0.1	170	15	20	7.2	3.5	1.0
13	54	Female	5.5	3.2	350	67	42	7.0	3.2	0.8
14	43	Male	22.5	11.8	143	22	143	6.6	2.1	0.46
15	57	Male	4.5	2.3	315	120	105	7.0	4.0	1.3
16	65	Male	0.7	0.2	199	19	22	6.3	3.6	1.3